



Department of Electrical & Electronic Engineering
Rajshahi University of Engineering & Technology

LAB Report

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Course Title : **Electronics Shop Practice**

Experiment No. : 05

Experiment Name : Design and Implementation of an Autonomous IR-Guided Line Follower Robot Using Arduino for Intelligent Line Tracking Applications

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Experiment No.: 05

Experiment Name: Design and Implementation of an Autonomous IR-Guided Line Follower Robot Using Arduino for Intelligent Line Tracking Applications

Objective:

1. To design and build an autonomous mobile robot capable of detecting and following a predefined path using Infrared (IR) sensors for line tracking.
2. To implement an Arduino-based control system that interprets IR sensor inputs and drives the robot's motors for real-time, precise navigation along the line.
3. To evaluate the robot's performance and tracking accuracy under various track conditions, and to optimize its behaviour through iterative software and hardware enhancements.

Theory:

A Line Follower Robot is an autonomous robotic system engineered to detect and follow a predetermined path—typically a black line on a white surface or vice versa. These types of robots are foundational in robotics education and have practical applications in automated material handling, warehouse logistics, industrial automation, and robotic competitions.

This project utilizes an Arduino Uno microcontroller, two IR sensors, and an L298N or L293D motor driver to construct a line-following robot that can make real-time path corrections. The robot uses the principle of infrared reflectivity to detect the contrast between the line and the background and adjusts its movements to stay on course.

Working Principle:

The operation of a line follower robot hinges on the light reflectivity principle. The IR sensor modules typically consist of:

- An IR LED (transmitter) that emits infrared light.
- A photodiode or phototransistor (receiver) that detects the reflected light.



Fig.05.1: View of an IR Sensor

Different surfaces reflect IR light differently:

- White surface: Reflects most of the IR light back to the sensor.
- Black surface: Absorbs IR light, resulting in minimal or no reflection.

Based on this contrast, the IR sensors provide digital or analog output to the Arduino, indicating the position of the robot relative to the path.

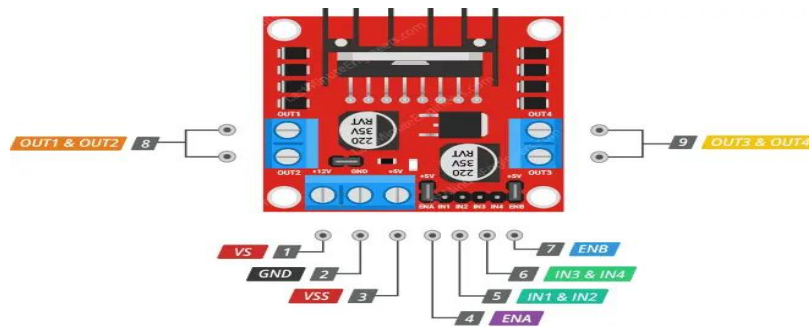


Fig.05.2: L298N Motor Driver

Control System and Logic:

The system operates using closed-loop feedback control:

1. IR sensors detect the current position relative to the line.
2. The Arduino Uno reads the sensor values and determines whether the robot is on track.
3. The motor driver receives control signals from the Arduino to adjust motor speeds and directions accordingly.

Typical Decision-Making Logic:

- Both sensors detect white: Robot moves forward.

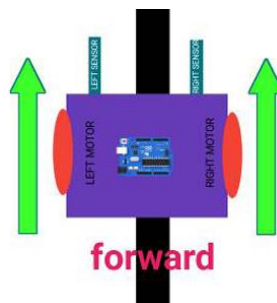


Fig.05.3: Condition When Motor Moves Forward

- Left sensor on black, right on white: Robot turns left.

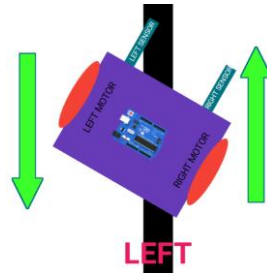


Fig.05.4: Condition When Motor Turns Left

- Right sensor on black, left on white: Robot turns right.

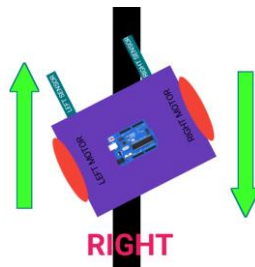


Fig.05.5: Condition When Motor Turns Right

- Both sensors on black (junction or error): Robot stops or takes a corrective turn.

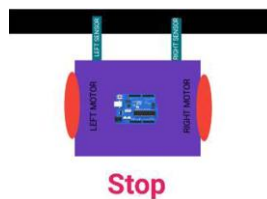


Fig.05.6: Condition When Motor Stops

The use of a differential drive system (where each motor is controlled independently) allows the robot to make sharp turns, pivot, and smoothly follow curves.

Advantages and Applications:

- Educational Value: Demonstrates fundamental principles of embedded systems, control loops, and sensor integration.
- Scalability: Can be enhanced with features like PID control, obstacle avoidance, or wireless monitoring.
- Practical Uses:
 - Assembly line automation
 - Smart delivery carts
 - Maze-solving robots
 - Competitive robotics

The Arduino-based line follower robot is a practical example of how microcontrollers, sensors, and motor control come together to create intelligent machines. By continuously analysing sensor data and adjusting motor behaviour, the robot can autonomously navigate complex paths with precision. This makes it an excellent project for beginners in robotics and a stepping stone toward more advanced autonomous systems.

Required Apparatus:

Table 5.1: Table For Required Apparatus

Sl. No.	Components	Specification	Quantity
1.	Arduino Uno	• ATmega328P • Operating Voltage: 5 V • Input Voltage: 7–12 V • Input Voltage: 6–20 V • Digital I/O Pins: 14 • PWM Digital I/O Pins: 6 (Pins 3, 5, 6, 9, 10, 11) • Analog Input Pins: 6 (A0 to A5) • DC Current per I/O Pin: 20 mA • DC Current for 3.3 V Pin: 50 mA • Flash Memory: 32 KB (ATmega328P) of which 0.5 KB used by bootloader • SRAM: 2 KB (ATmega328P) • EEPROM: 1 KB (ATmega328P) • Clock Speed: 16 MHz • USB Connection: Type B • LED Built-in: Pin 13	1

2.	L298N Motor Driver	• Driver Model: L298N 2A • Driver Chip: Double H Bridge L298N • Motor Supply Voltage (Maximum): 46V • Motor Supply Current (Maximum): 2A • Logic Voltage: 5V • Driver Voltage: 5-35V • Driver Current: 2A • Logical Current: 0-36mA • Maximum Power (W): 25W • Current Sense for each motor • Heatsink for better performance • Power-On LED indicator	1
3.	Gear Motors	• Operating Voltage: 6V • Shaft Length: 8.5mm • Shaft diameter: 5.5mm • Rated Speed: 100-150RPM • No Load Current: 40-180mA • Rated Torque: 0.35 Kg-cm • Dimensions (L×W×H) in mm: 65×23×27	2
4.	Wheel	-	2
5.	Battery	9V	1
6.	Jumper Wires	-	-
7.	IDE Software	Arduino IDE	-
8.	Breadboard	-	1
9.	Robot Chassis	-	1
10.	Glue Gun	-	1

Circuit Diagram:

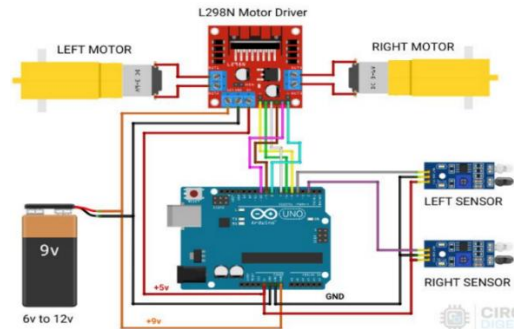


Fig.05.7: Circuit Diagram of Line Follower Robot

Experimental Setup:

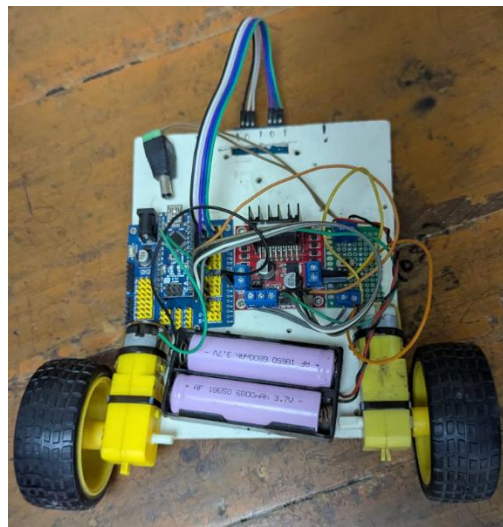


Fig.05.8: Experimental setup of the Line Follower Robot

Code:

<pre>#define enA 5 #define in1 6 #define in2 7 #define in3 9 #define in4 10 #define enB 8 #define R_S 4 #define L_S 2 void setup(){ pinMode(R_S, INPUT); pinMode(L_S, INPUT); pinMode(enA, OUTPUT); pinMode(in1, OUTPUT); pinMode(in2, OUTPUT); pinMode(in3, OUTPUT); pinMode(in4, OUTPUT); pinMode(enB, OUTPUT); digitalWrite(enA, HIGH); digitalWrite(enB, HIGH); delay(1000); } void loop(){</pre>	<pre>void forward(){ digitalWrite(in1, HIGH); digitalWrite(in2, LOW); digitalWrite(in3, LOW); digitalWrite(in4, HIGH); } void turnRight(){ digitalWrite(in1, LOW); digitalWrite(in2, HIGH); digitalWrite(in3, LOW); digitalWrite(in4, HIGH); } void turnLeft(){ digitalWrite(in1, HIGH); digitalWrite(in2, LOW); digitalWrite(in3, HIGH); digitalWrite(in4, LOW);</pre>
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<pre> if((digitalRead(R_S) == 0)&&(digitalRead(L_S) == 0)){forward();} if((digitalRead(R_S) == 1)&&(digitalRead(L_S) == 0)){turnRight();} if((digitalRead(R_S) == 0)&&(digitalRead(L_S) == 1)){turnLeft();} if((digitalRead(R_S) == 1)&&(digitalRead(L_S) == 1)){Stop();} } </pre>	<pre> } void Stop(){ digitalWrite(in1, LOW); digitalWrite(in2, LOW); digitalWrite(in3, LOW); digitalWrite(in4, LOW); } </pre>
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Experimental Procedure:

1. Component Collection and Inspection
 - Gather all the required components:
 - Arduino Uno
 - L298N Motor Driver
 - 2 IR Sensors
 - 2 DC Gear Motors with wheels
 - Robot chassis
 - Battery pack (2×18650 Li-ion, 7.4V)
 - Jumper wires, on/off switch
 - Ensure all components are in working condition before assembly.
2. Chassis Assembly
 - Mount the motors onto the chassis using screws or mounting brackets.
 - Fix the wheels to the motor shafts.
 - Attach the caster wheel for balance.
3. Motor and Driver Wiring
 - Connect both DC motors to the motor output terminals (OUT1–OUT4) of the L298 driver.
 - Connect the input pins (IN1–IN4) of the driver to digital pins on the Arduino as per your code.
 - Connect the Enable pins (EN A and EN B) to PWM-capable Arduino pins for speed control.
 - Power the motor driver using the battery pack (7.4V to 12V).
 - Ground of battery, Arduino, and motor driver should be connected in common.
4. IR Sensor Connections
 - Fix two IR sensors at the front of the robot, spaced properly to detect the black line underneath.
 - Connect the VCC and GND of both sensors to 5V and GND on Arduino.
 - Connect their digital output pins to the Arduino's input pins defined in the code
5. Uploading the Arduino Code
 - Open the Arduino IDE on your computer.
 - Connect Arduino Uno to your PC via USB.
 - Paste the finalized code (without comments or with, as per need) into the IDE.
 - Select the correct board and COM port, then upload the code.
6. Power Supply Integration

- Disconnect USB and power the Arduino with the battery pack via the VIN pin or barrel jack.
 - Ensure a power switch is installed between the battery and motor driver for easy control.
7. Testing the Robot
- Place the robot on a black line over a white background.
 - Turn on the switch and observe behaviour:
 - Robot should move forward when both sensors detect white.
 - It should turn right or left when one sensor detects black.
 - It should stop or correct when both detect black.
8. Calibration and Adjustments
- Adjust sensor positions to improve accuracy.
 - Modify sensor threshold values or motor speed in the code if performance is unstable.
 - Ensure the line width and contrast are appropriate for the IR sensors to function reliably.
9. Final Evaluation
- Test on various line tracks including straight lines, curves, and junctions.
 - Document robot response, turning accuracy, and line-tracking efficiency.

Discussions:

In this lab, creating a basic line following robot using the L293D motor driver IC and Arduino microcontroller was the main objective. After trying many times, the experiment was a success. At first, the robot couldn't advance properly towards the black line due to less friction of the wheels. But at last, after fixing the wheels, the experiment was a success.

Justification of CO2 and PO(j):

This design uses IR sensor inputs to control motor movement based on conditions: moving forward, turning, or stopping. Motors are safely enabled with a delay, and functions are clearly defined for each action.

Conclusion:.

The IR-guided line follower robot was implemented using Arduino and basic electronic components. It could follow a black line by responding to sensor inputs in real time. This project demonstrated the fundamentals of intelligent line tracking using simple logic. It provided hands on experience in robotics, sensor integration, and programming.

Reference:

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2. <https://lastminuteengineers.com/l298n-dc-stepper-driver-arduino-tutorial/>
3. <https://circuitdigest.com/microcontroller-projects/arduino-uno-line-follower-robot>

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